

A Hormone-Driven Epigenetic Mechanism for Adaptation in Autonomous Robots

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Abstract—Different epigenetic mechanisms provide biological organisms with the ability to adjust their physiology and/or morphology and adapt to a wide range of challenges posed by their environments. In particular, one type of epigenetic process, in which hormone concentrations are linked to the regulation of hormone receptors, has been shown to have implications for behavioral development. In this paper, taking inspiration from these biological processes, we investigate whether an epigenetic model based on the concept of hormonal regulation of receptors can provide a similarly robust and general adaptive mechanism for autonomous robots. We have implemented our model using a Koala robot, and tested it in a series of experiments in six different environments with varying challenges to negotiate. Our results, including the emergence of varied behaviors that permit the robot to exploit its current environment, demonstrate the potential of our epigenetic model as a general mechanism for adaptation in autonomous robots.

Index Terms—Allostasis, autonomous robot, emergent behavior, epigenetic development, hormonal modulation.

I. INTRODUCTION

ONE OF the biggest challenges in autonomous robot research is developing controllers that can adapt to the unstructured, novel, and dynamic conditions found in the real world. As pointed out by Krichmar [1], while significant advances have been seen in the capabilities of autonomous systems, controllers tend to be based on mechanisms that are developed to operate in highly task-specific and relatively controlled environments. While these models have proven to foster adaptation in their specific environments, questions must be asked about their ability to recreate this performance in less controlled settings, where even small changes in conditions can have significant, and often unpredictable, implications for the behavior of robots [2]–[4]. To operate and adapt autonomously in real world environments, robot controllers must not only be robust but also have flexible and general adaptation capabilities, allowing a robot to adapt to different environmental conditions changing across

environments and over time. In biological organisms, epigenetic processes [5]–[8] provide one such adaptation mechanism. Drawing inspiration from development in biological organisms, we claim that an epigenetic process in which environmental stimuli are able to affect hormone receptors could potentially provide a very useful general mechanism for adaptation in autonomous robots.

In this paper, we present a simple model of such an epigenetic mechanism, in which the sensitivity of receptors for simulated hormones is directly regulated by the hormone level. In our model, we use varying simulated hormone concentrations, secreted as a function of the interactions of a robot with its environment, not only to “rewire” different neural or physiological functions as in, e.g., [9]–[11] but also to regulate their receptors. This regulation functions in a similar manner to a positive feedback loop. Higher concentrations of a hormone lead to upregulation (increased sensitivity) of a receptor and lower concentrations cause downregulation (reduced sensitivity). Here, we investigate the adaptability of our epigenetic mechanism to a range of environmental challenges through allostatic processes [12] and demonstrate, in a series of experiments, how our system can lead initially identical robots (i.e., robots with the same initial internal parameter values and with the same speed potential) to develop different behavioral phenotypes well adapted to different environmental conditions, as a consequence of the changes brought about by, and in the epigenetic mechanism in its interactions with, the environment. In our model, no predefined behavioral repertoires or hand-crafted behaviors are used. Instead, all behaviors observed and discussed in this paper emerged as a result of both short-term (hormonal) and long-term (epigenetic) modulation of the perception and actuators of the robot.

The work presented here stems from, but expands very significantly, an earlier conference publication [13]. Here, we provide a much deeper investigation by both adding new experimental conditions, analyzing the adaptive mechanisms of the robot more rigorously, and exploring the effect of experiential diversity on development in much greater detail. We refer the reader to [14] for a comprehensive look into the background of the approach.

The remainder of this paper is organized as follows. In Section II, we introduce the different aspects of our model. In Section III, we present the different experiments carried out to test our model and the results obtained. Finally, we draw some conclusions in Section IV.

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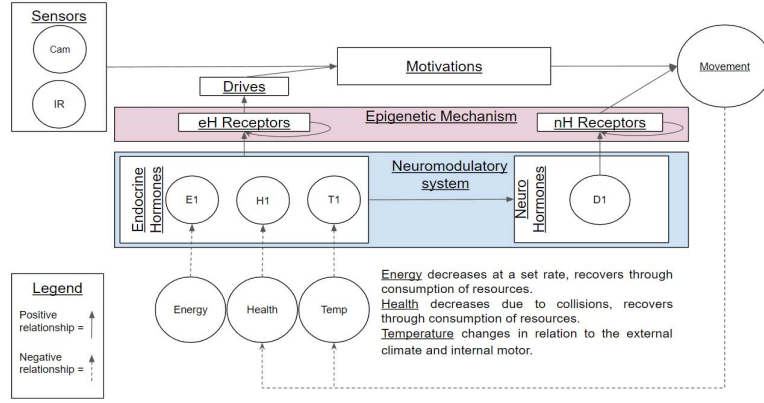


Fig. 1. Complete robot architecture with its three interacting “systems”: basic architecture, neuromodulatory system, and epigenetic mechanism.

II. ROBOT MODEL

Our epigenetic robot architecture can be broken down into three systems, designed to build on each other incrementally.

- 1) *Basic Architecture*: The first system handles basic functions such as drives, motivational state, and movement (Section II-A).
- 2) *Neuromodulatory System*: The second system modulates the basic architecture through the secretion and decay of four different simulated hormones (Section II-B).
- 3) *Epigenetic Mechanism*: The third system implements a simple epigenetic mechanism, which controls the sensitivity of the hormone receptors of the second system (Section II-C).

Since this architecture is designed incrementally, we can implement robots driven by the first system only (which we will call “Basic robot” in this paper), by a combination of the first and the second systems (called “hormone-modulated robot”), or by the three interacting systems (called “epigenetic robot”). The full (epigenetic) robot architecture, containing the three systems, is depicted in Fig. 1. This incremental design of the robot architecture permits a methodical, incremental testing, and analysis of the functionalities and benefits (in terms of adaptation) that each subsequent addition brings to the robot. To this end, we tested each of the three robot models in the same six open-field environments posing different challenges, as we will explain in Section III.

A. Basic Architecture

The basic behavior of the robot is driven by a combination of internal needs and external stimuli. The “basic architecture” is built around a motivational system that generates continuous behavior, running continuously in “action selection” loops (or time steps) of 62.5 ms (16 loops per second).

1) *Motivational System*: Following Cañamero’s long-standing approach [15], our basic robot has a number of internal needs arising from a simulated physiology of homeostatically controlled survival-related variables. Our robot has three simulated survival-related variables: 1) **energy**; 2) **health**; and 3) **temperature** (all with values between 0 and 100), which, to survive, it must maintain within a predefined range.

The variable **energy** acts as a simulated battery that the robot can sense internally. This simulated battery has a maximum charge of 100 units (ideal value), a “fatal limit” (the robot “dies” if this is reached) of 0, and loses one unit of charge per second. The robot is able to recharge this battery by moving within a 2 cm range of an “energy resource”—a pink ball.

The variable **health**, a simulated measure of physical integrity, also has an ideal value of 100 units and a fatal limit of 0. Its actual value decreases with physical contact, which our robot detects with its 16 infrared sensors (IR), according to

$$\Delta\text{Health} = \sum_i \begin{cases} -\frac{\text{Ir}_i}{\text{IrMax}} & \text{if } \text{Ir}_i > \text{IrMax} \times 0.95 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where Ir_i is the value of the IR i , and IrMax is the maximum sensor value, corresponding to an object touching the sensors. Hence, if all 16 IR detected an object in contact with the robot, then the robot would lose 16 health units per action loop. Like energy, this internal need can be recovered by consuming specific resources, in this case blue colored balls.

Finally, the (internal) variable **temperature** depends on the current wheel speed of the robot and the simulated temperature of the external environmental, which we call **climate**. It has an ideal value of 0 and a fatal limit of 100; therefore, the error of this variable is not by deficit but by excess, and coincides with the actual internal temperature. The temperature changes according to

$$\Delta\text{Temperature} = \frac{|\text{sp}|}{10} \times \text{Climate} - (\text{Temperature} \times 0.05) \quad (2)$$

where $|\text{sp}|$ is the absolute value of the wheel speed (measured in rotations per action loop) and the value 10 is a predetermined constant to regulate temperature gain.

Following a homeostatic model, the discrepancy between the actual and ideal values of an internal variable produces an error and a tendency to correct that error—a *drive*. The robot has two explicit drives: a tendency to correct energy deficits, i.e., increase energy levels, (drive_e), which is satisfied by consuming “energy” (pink balls), and a tendency to repair (increase the level of) health (drive_h), which is satisfied by consuming “medicine” (blue balls). The need to decrease temperature is indirectly satisfied by reducing movement; this

TABLE I

RELATIONSHIP BETWEEN DRIVES AND ENVIRONMENTAL CUES. THE NUMBERS SHOWN ARE THE SCALING FACTORS USED IN (4)

Drive	Environmental Cues		
	Energy Resource	Health Resource	Solid Object
Replenish energy	10	0	0
Repair health	0	10	-1

means that it can interfere with the satisfaction of the other drives, which involve movement of the robot. Therefore, these interactions need to be taken into account when calculating the intensity of the drives:

$$\text{Drive}_i = \frac{\text{Deficit}_i}{\text{Error}_i} \quad (3)$$

where Drive_i is the need to recover the homeostatic variable i ($=e$ or h), Deficit_i is the deficit of the current homeostatic variable i , and Error_i is the temperature error.

The drives affect the robot's perception of three different environmental cues using the scaling factors shown in Table I. In this paper, as the relationship between drive and cue is fixed, the robot's perception of different cues can only be modulated by its internal state, allowing us to focus on the effects of the epigenetic model. However, in [16] we have integrated the epigenetic model into a developmental learning system permitting the robot to learn the affordances of different objects, i.e., to learn how it can interact with them, or the effect of its interactions.

In addition to its internal needs, the motivational system of the robot takes into account the external stimuli or environmental "cues" relevant to the satisfaction of its different needs (Table I) to decide on the appropriate course of action. Departing from previous models that used discrete motivations (e.g., [9], [15], [17], and [18]), in this architecture the robot has a global motivational state that will make it move at a particular speed and in a particular direction—one of eight directions around the robot, determined in body-coordinates. The motivational state of the robot (a vector) is calculated as follows:

$$\text{MotState}_j = \sum_{i \in \{e, h\}} \sum_c s_{i,c} (\text{Drive}_i \times \text{Cue}_{c,j}) - \text{Avoid}_j \quad (4)$$

where j is one of eight directions around the body of the robot, $s_{i,c}$ are the scaling factors in Table I, $\text{Cue}_{c,j}$ is the magnitude of an environmental cue of type c in direction j , and Avoid_j is given by

$$\text{Avoid}_j = \begin{cases} \text{Ir}_j & \text{if } \text{IrMax} - \text{Ir}_j < \text{Drive}_h \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

2) *Behavior Generation*: The motivational state influences the generation of the behavior (motor action) to be executed. Unlike the majority of related motivational robot architectures previously developed in our group, which use discrete behaviors (e.g., [9], [15], [17], and [18]), our robot uses a continuous sensory-motor mapping model, more in vein with models such as [19]. For example, rather than having an explicit "pushing" behavior, the robot may push an object when its "desire" to

move forward outweighs its desire to avoid the object. Our model thus calculates the motion of the robot rather than selecting specific predefined behaviors. Motion is calculated independently for each wheel and the overall movement (speed and direction) results from the simultaneous movement of both wheels. The speed of the wheels, and therefore, the observable behavior, is calculated by

$$\text{WheelSpeed}_k = \sum_{j=0}^7 (\text{MotState}_j \times \text{Set}_{k,j}) \quad (6)$$

where WheelSpeed_k is the speed of the left or right wheel ($k = 0$ or 1), and $\text{Set}_{k,j}$ are vectors $(-10, -10, -5, -3, 1, 3, 5, 10)$ if $k = 0$ and $(-10, 10, 5, 3, 1, -3, -5, -10)$ if $k = 1$, to translate the eight-directional motivational state vector into a coherent motor action.

As shown in (4) and (6), the robot's movement is thus driven by both its internal state and external¹ stimuli. When the robot's internal needs are satisfied, it will engage in an avoidance-like behavior and try to maintain a distance between itself and the objects perceived, stopping if it finds an empty² area. In contrast, when the robot needs to replenish health or energy but no resources are detected, it will move forward (avoiding obstacles) looking for them.

B. Neuromodulatory System

Our neuromodulatory system implements a hormonal system that modulates the basic motivational robot architecture. Although such simple motivational models can be effective behavior selection systems in simple environments, they tend to perform poorly in more complex and dynamic environments. A potential solution to this problem is found in biological systems in which motivations do not come directly from homeostatic errors. Rather, hormone secretion derived from homeostatic errors are behind the onset of motivation by providing a signal of the error [20] and the motivational value of environmental cues [21]. In our previous work, we have applied hormonal modulation to both the perception of external environmental cues and of internal errors (e.g., [9], [15], and [18]) as well as to the internal elements of the architecture linking perception and actuation (e.g., [22] and [23]). In this paper, (simulated) hormones secreted from (simulated) glands modulate different sensors and actuators of the robot. These hormones are secreted in relation to changes in the robot's internal and external environment. Our model implements four hormones of two different types: 1) endocrine hormones (Eh) and 2) neurohormones (Nh), as shown in Table II.

1) *Endocrine Hormones*: Drawing on biological systems, our Eh hormones are implemented with the primary purpose of helping to maintain internal homeostasis. The Eh group

¹As we mentioned above and shown in Table I, the robot is able to detect three types of environmental cues that affect its motivational state and ultimately its behavior. For example, if the robot sees an energy resource and the intensity of its energy drive is 2, then its motivation to move toward the resource would be equal to $20 \times \text{Set}_{k,j}$.

²Due to the range of the Koala robot's IR sensors, an "empty area" roughly corresponds to a gap of 5 cm between the robot's IR sensors and any object.

TABLE II
DIFFERENT HORMONES USED IN OUR ARCHITECTURE

Hormone	Type	Secretion trigger	Regulates
E1	Eh	Energy error (deficit)	Energy recovery
H1	Eh	Health error (deficit)	Health recovery
T1	Eh	Internal temperature error (excess)	Speed
D1	Nh	Visual cues and Eh concentration	“Dominant / aggressive” behavior

consists of three hormones, each one associated with one of the three homeostatic variables. These hormones are secreted as a function of current homeostatic errors

$$\text{EhSecretion}_i = \psi_i \times \text{Error}_i \quad (7)$$

where i is one of the three homeostatic variables, and ψ_i is a constant value to regulate secretion.

2) *Neurohormones*: We have implemented a single Nh: D1. This hormone facilitates what can be described as “dominant” or potentially “aggressive” behavior, and can be thought of as being somewhat similar to testosterone in mammals. This dominant behavior is achieved by the combination of two effects of Nh: 1) changing the wheel speed, as we will see in (11), and 2) reducing the tendency to avoid objects by modulating the perception of environmental cues that are associated with negative stimuli such as collisions, overheating or other actions which move the robot away from homeostatic balance (10). This system shares some similarities with [9], where a hormone suppressed perceptual feedback from the robot’s bumpers (touch sensors), causing the robot to push competitors away as a side effect. However, in our model the Nh hormones have the effect of suppressing all negative stimuli. In addition, in our model the strength of suppression of negative stimuli is correlated with the concentration of hormones present in the system, as in [9], rather than with internal deficits, providing a simple form of “affective memory.” For example, a robot with a high D1 level that detects a desired resource will move directly toward it ignoring and pushing aside any obstacles in the way, disregarding the potential damage from collisions. In contrast, a robot in the same situation but with a low D1 level would instead move around obstacles to reach the desired location. Another example is the ability of the Nh to modulate the robot’s speed by suppressing the drive to slow down due to overheating.

The amount of secretion of the Nh is determined by the intensity of all the internal drives, as given by

$$\text{NhSecretion} = \sum_i \text{Drive}_i. \quad (8)$$

3) *Hormone Receptors*: Once secreted, both types of hormone, Eh and Nh, enter the virtual bloodstream that surrounds the architecture and persist for a random number of action selection loops (within a fixed range) before decay of that particular secretion occurs. Hormone receptors can detect the different concentration of these two types of hormones. These receptors are then able to signal for the appropriate

modulation in response to internal or environmental change. This modulation is achieved by modifying (3) to

$$\text{Drive}_i = \frac{\text{Sens}_i \times \text{Eh}_i}{\text{Sens}_i \times \text{Eh}_i} \quad (9)$$

where t is temperature, $i \neq t$ is one of the other two survival-related variables, and Sens_i is the sensitivity of the receptor to the level of the corresponding hormone. In the epigenetic robot (seen Section II-C), Sens_i is modulated by the epigenetic mechanism, but in the hormone-modulated robot, Sens_i is a constant = 1.

The suppression of negative stimuli by the Nh is achieved by modifying (5) and (6), respectively, to

$$\text{Avoid}_j = \begin{cases} \frac{\text{Ir}_j}{\text{Nh}} & \text{if } \text{IrMax} - \text{Ir}_j < \text{Drive}_h \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

$$\text{WheelSpeed}_k = \sum_{j=0}^7 (\text{MotState}_j \times \text{Set}_{k,j}) \text{Sens}_{\text{Nh}} \times \text{Nh}. \quad (11)$$

C. Epigenetic System

The final system of the model introduces an epigenetic adaptation mechanism into the architecture. Taking inspiration from biological studies (see [5], [6], [8] for an overview) in which hormone concentrations are shown to correlate with the triggering of varying epigenetic mechanisms that influence the expression of hormone receptors, we use simulated hormones in our model to regulate the robot’s hormone receptors. High concentration of a hormone leads to the upregulation of the receptors (i.e., an increased number of receptors) and lower concentrations cause downregulation (i.e., a lowered number of receptors). As we show in this paper, changes in regulation of receptors lead to the emergence of persistent behaviors relevant to the robot’s current environment. These behaviors can of course still be modulated by variations in hormone concentration levels.

In this paper, we have not tried to implement an explicit model of the epigenetic process (i.e., an explicit model of DNA methylation), but rather we have modeled the outcome of the process by updating the previous value of the Sens_i in (9) as follows:

$$\text{Sens}_i(t+1) = \text{Sens}_i(t) \frac{\text{Eh}_i}{\sigma} \quad (12)$$

where σ is a predetermined constant value that regulates the speed of the epigenetic process, $\text{Sens}_i(t)$ is the sensitivity of the receptor for hormone i at time step t , and Eh_i is the hormone concentration; hence, the hormone influences the upward or downward regulation of its own receptors.

III. EXPERIMENTS

To assess the ability of the epigenetic robot to adapt to different environmental challenges, we tested it in six different environments, where we also tested the basic and neuromodulated robots for comparison. The purpose of these experiments was to investigate the adaptability of our epigenetic mechanism to a range of environmental challenges through allostatic processes [12], and more specifically, how our system can

TABLE III
OVERVIEW OF THE DIFFERENT EXPERIMENTS AND THE CHALLENGES THEY POSED TO THE ROBOT

Experiment	Challenge
1) Basic Environment	An open environment with no significant challenges, used as a baseline for comparison with more complex environments. Two items of each resource were present at all times.
2) Light Objects	The environment contains light objects that the robot needs to push past in order to reach resources. As pushing results in a loss of health, this creates a trade-off for the robot between maintaining its energy or its health.
3) Moving Resources	The energy resources move in a constant pattern at a set speed, attempting to simulate a simple prey. As they move at the same speed as the robot, the robot cannot simply chase after them to feed.
4) Dynamic Climate	The temperature of the environment changes through the experiment in a day-night cycle, with hotter temperatures during the “day” and slowly cooling down at “night”. This challenges the robot to satisfy drives during cooler “night” periods so that it can reduce movement during the day and not overheat.
5) Uneven Resources	Three sets of 5 runs with uneven resource distribution in each set of runs. Each set placed stress on a specific homeostatic variables (e.g., one set of runs would be in an environment with a reduced number of energy resources). This environment challenges the robot to adapt to prioritize less abundant/harder-to-maintain needs.
6) Temporal Dynamics	The resources are only available for limited periods of time. This forces the robot to take opportunities to recover deficits when the resources appear. For the first five sets of runs, the energy source would be present for 30 seconds per minute. In the second five sets of runs the duration was reduced to 20 seconds. In the final five sets of runs the resource was only accessible for 10 seconds of every minute. As only one type of resource is affected the robot must prioritize this one over the others.

allow initially identical robots (with the same initial internal parameter values and speed potential) to develop different behavioral phenotypes adapted to different environments as a consequence of the changes brought about by, and in, the epigenetic mechanism in its interactions with the environment. All the adaptations took place as a result of such dynamics of interactions, without the need for programming or tuning the robot to operate in a specific environment.

Adaptation was assessed in terms of how well the internal physiological variables were maintained, quantitatively measured in terms of “comfort level” [9] (Figs. 4–7, 9, and 10), which we calculated as the normalized mean of the robot’s Energy, Health, and $100 - \text{Temperature}$.

These experiments were carried out using the medium-sized wheeled Koala-II robot.³ In addition, a webcam was added to the robot to allow for simple color-based vision. Computations were performed on a laptop fitted on top of the robot. Each action selection loop lasted 62.5 ms, equivalent to 16 loops per second.

The experiments were performed inside a $2\text{ m} \times 2\text{ m}$ arena with wooden borders, shown in Fig. 2. Each of our first four experiments consisted of ten runs for each robot architecture, with a duration of 10 000 steps (action selection loops) each, or approximately 10 min at a rate of 16 steps per second. The last two experiments involved 15 runs of 10 000 steps for each architecture (five runs in each of the three environmental sub-conditions – see Sections III-E and III-F). In each experiment, the robots had access to at least one item of each resource, with the exact number and environmental challenges changing across experimental sets. If at any point one of the robot’s homeostatic needs passed its fatal limit, it “died” and the run would end. In these case homeostatic values are recorded as 0 for any time step occurring after the “death”.

Table III provides an overview of each experiment, Table IV shows the physiological changes that the epigenetic robot underwent in terms of receptor regulation in the six



Fig. 2. Koala robot used in our experiments. Additionally, the light movable objects from experiment 2 and a moving “prey” resource from experiment 3 can be seen at the top and left of the figure, respectively.

experiments, and Fig. 3 shows the evolution of the sensitivities of the hormone receptors over the course of the experiments and observed behaviors.

A. Results of Experiment 1: Basic Environment

In this simple environment all three robot architectures were able to maintain homeostatic balance (see Fig. 4). However, while the three models were similar in terms of homeostatic maintenance, they varied in their behavior.

The basic architecture and hormone-modulated robots behaved as expected: homeostatic errors would cause modulation of the wheels [see (6), (11)] and motivate the robots to move toward and consume the appropriate resources. In contrast, the epigenetic robot stayed close to an energy resource (all located in a corner) at all times even when not consuming it, either stopping nearby or tightly circling it. This behavior occurred due to a downregulation of the H1 and T1 hormone receptors shown in Table IV and Fig. 3. Limited presence of the H1 and T1 hormone due to the simplicity of the environment (few obstacles and limited required movement) resulted in this change in expression. As the energy decreased at a set rate, the E1 receptor remained constant. By staying stationary, this robot was able to almost entirely eliminate the other drives. As can be seen in Fig. 4, while results in terms of

³<http://www.k-team.com/mobile-robotics-products/old-products/koala>

TABLE IV
RECEPTOR REGULATION OF THE EPIGENETIC ROBOT. AN UNREGULATED RECEPTOR WOULD HAVE A VALUE OF 1, UPREGULATED >1 , DOWNREGULATED <1

Experiment	E1 Receptor	H1 Receptor	T1 Receptor	D1 Receptor
1) Basic Environment	1.01	0.69	0.86	0.41
2) Light objects	1.21	1.34	1.02	1.29
3) Moving Objects	1.43	1.13	1.56	1.33
4) Dynamic climate	1.31	1.03	1.78	0.91
5) Uneven Resources set 1	0.89	1.14	0.91	0.86
set 2	1.25	0.72	1.07	0.82
set 3	1.06	1.01	1.22	0.80
6) Temporal Dynamics	1.54	0.86	0.94	1.21

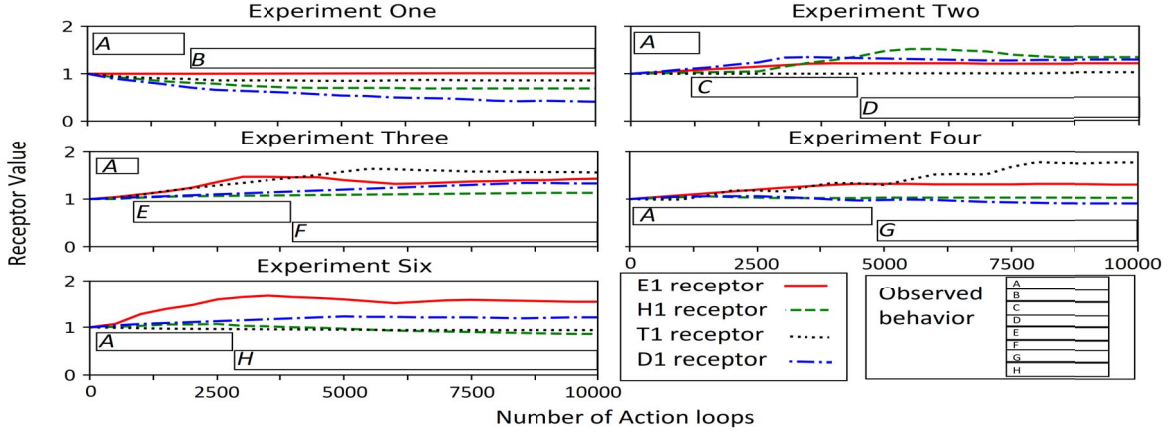


Fig. 3. Graphs showing the changes in sensitivities of the hormone receptors in the epigenetic robot over time. Letters indicate the occurrence of observed robot behaviors that will be discussed in the experiments (Sections III-A–III-F). A) Simple reactive behavior. B) Due to downregulation, the robot is attracted to only one aspect of the environment using a guarding behavior. C) Reduced object avoidance due to upregulation of D1, robot pushes past obstacles. D) Pushing behavior refined, as a result of upregulation of H1. The robot now repairs before attempting to push. E) Robot chases after resources, slowly gaining speed as motivation to reach them increases until it reaches them or overheats and stops. F) Upregulation of T1 means the robot using an ambush strategy, staying still until motivation to move to a resource increases (due to close proximity) at which point upregulation of E1/D1 ensures the robot moves at full speed. G) Upregulation of T1 results in a “hibernation” behavior, exploiting favorable weather condition to recover homeostatic deficits before returning to a dormant state. H) Rapid upregulation of E1 leads the robot to a state of extreme sensitivity to opportunities to recover energy. This heightened sensitivity will override any other needs, allowing the robot to take advantage of the limited windows of opportunity to recover energy. Upregulation of D1 means that the robot will move at higher speed toward the resources.

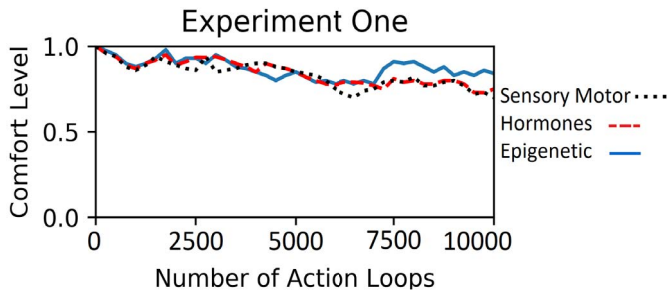


Fig. 4. Results of the first experiment (basic environment).

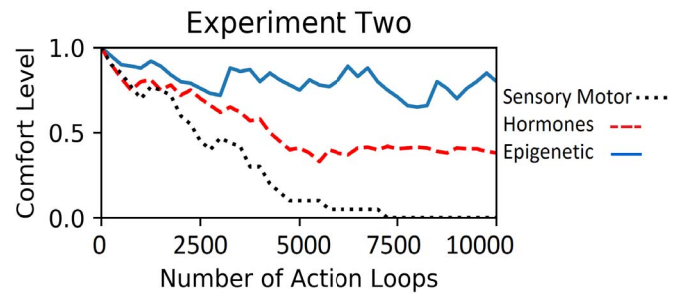


Fig. 5. Results of the second experiment (light objects).

adaptation of all robots (as measured by comfort levels) are close, the epigenetic robot maintained slightly higher overall values particularly in the latter stages, supporting idea that in this case lack of movement was adaptive.

B. Results of Experiment 2: Movable Objects

In this scenario, the nonepigenetic robots performed poorly (see Fig. 5): the basic architecture robot died in all runs and,

while the hormone-modulated model performed better, with only three deaths overall, its performance was still poor. The inability of these architectures to adapt to the harsher environment stemmed from two causes. First, they often failed to maintain sufficient health to push past obstacles to reach the resources (i.e., health deficits led to increased collision avoidance). Second, they were unable to adapt to the longer time needed to find a resource due to the need to navigate obstacles.

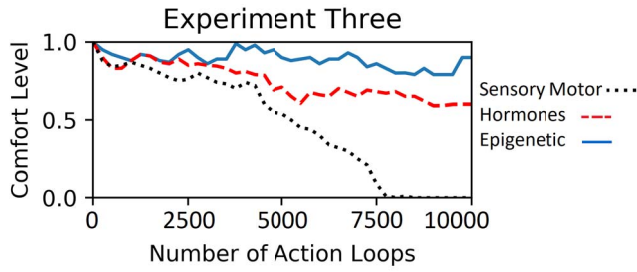


Fig. 6. Results of the third experiment (moving resources).

These issues resulted in both low energy and poor health, whereby robots died due to either running out of energy, or from impact of pushing past objects.

In comparison, the epigenetic robot was able to adapt to the new environment due to a chain reaction of internal changes (see Table IV). First, perception of resources that were visible but inaccessible led to persistent high levels of the D1 hormones and an upregulation of the D1 receptor. The increase in D1 sensitivity made this robot more likely to push past obstacles rather than going around. This behavior resulted in increased health deficits, and subsequently, increased concentrations of the H1 hormone and the upregulation of the H1 receptor. This increased the motivation to replenish health after pushing past an object, and hence led to maintenance of high levels of health. These epigenetic changes ensured that this robot could safely push past objects when needed.

We also observed differences in the strength with which the robots “pushed” objects. The way in which the robots changed the force of their push was by driving faster or slower into an object.⁴ The epigenetic robot adapted to a more forceful pushing behavior as a result of upregulation of D1 due to environmental pressures. These intense pushes resulted in lower health loss from purposeful collisions due to minimising contact exposure. Forceful pushes also increased the likelihood of the epigenetic robot to fully complete the pushing behavior, pushing completely past an object rather than giving up and moving away part way through. In this experiment, the average speed of the epigenetic robot while trying to push was 230 mm/s, in comparison to 150 mm/s for the hormone-modulated and 60 mm/s for the basic architecture robots.

C. Results of Experiment 3: Moving Resources

In this environment the resources moved in regular patterns at the default speed of the robots, simulating a simple prey. Due to their similar speeds, the robots could not simply chase a resource to reach it and needed to develop a “hunting” behavior to avoid drawn-out “chases” that could make them overheat critically. As seen in Fig. 6, the epigenetic robot’s behavior proved most suitable. This robot developed an “ambush-like” strategy, where it would remain sedentary until a needed resource passed nearby, at which point it would “pounce,” giving chase at full speed. This behavior emerged due the following mechanisms: overheating from

⁴The speed of the push of the robots depended on the intensity of the temperature drive (T1), which decreased speed, and D1 hormone concentration, which increased speed [see (9) and (11), respectively].

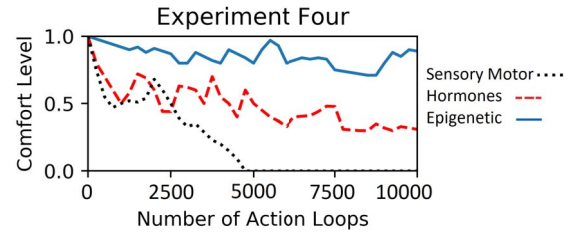


Fig. 7. Results of the fourth experiment (dynamic climate).

initial chases resulted in significant upregulation of the T1 receptor increasing sensitivity to its temperature and subsequently in suppressed movement, which led to a low internal temperature. The robot was able to overcome this suppression, and thus move, either with very high E1/H1 concentrations or if a resource came within close distance.⁵ Due to the finely balanced internal parameters and the robot’s low internal temperature, once the motivation to move developed the robot was able to move at full speed for short bursts. Due to the speed of the epigenetic robot when it gave chase, the resource was often pinned against a wall, making it easier for the robot to feed on it; this also reduced the length of the chases, which lasted only around 4 s and were successful 87% of the time.

The hormone-modulated model also had a high success rate of 72% but showed longer chases of around 14 s on average. Compared with the epigenetic robot, the hormone-modulated robot engaged in long chases, often catching the resource when it turned in a corner. During these chases this robot’s speed would increase the closer it got to the resource (more intense stimulus) and the longer the chase went on (increased D1 concentration). Failure here was always due to the fact that the robot overheated before it reached the resource and needed to suspend or slow down movement in order to reduce its temperature.

Finally, the basic architecture robot only had a success rate of 13%, with an average chase taking 12 s. This robot would follow a needed resource; however, due to the lack of the D1 hormone it could rarely catch up with a resource. Unless the robot happened to be near one of the end points where the resource stopped, it would abandon the chase due to overheating. The inability to catch resources meant that this robot constantly had critically low levels of the survival-related internal variables and in all cases died before the end of the experiment. Due to its low success rate, the basic architecture robot also spent a larger amount of time engaging in chases—67% of its time in comparison to 28% for the hormone-modulated robot and just 7% for the epigenetic robot.

D. Results of Experiment 4: Dynamic Climate Cycle

As can be seen in Fig. 7, the basic architecture robot performed poorly in this environment as it died before the end of the second climate cycle, with causes of death evenly split between lack of energy and overheating. Simply put, this robot did not adapt to take advantage of the colder period, and still had energy or health deficits during the hot period.

⁵By the end of a run, assuming base levels of the E1 hormone, this distance was around 12 cm, with higher concentrations increasing it.

This resulted in a lose-lose situation: either moving to replenish energy or health and dying of overheating, or staying still and running out of energy.

The hormone-modulated robot fared better. While it was still unable to fully take advantage of the cold periods, a slow-paced moving behavior emerged during the hot periods due to a combination of high T1 and medium levels of E1 and H1. However, even with this behavior, three deaths were still recorded due to energy loss.

Finally, the epigenetic robot developed two contrasting behaviors in order to survive the periods of high ambient temperature. In some (seven) runs, it developed a hibernation behavior: the robot became highly attracted to the different resources during the cold period and thus would fully replenish any deficits before laying dormant during the hot periods. This hibernation behavior emerged due to upregulation of the three Eh receptors (Table IV). In particular, upregulation of the T1 receptor (due the heat during the day) resulted in the robot completely suppressing movement during these periods. As the robot could not move during the day, in the early stages deficits in health or energy could not be corrected during this period, resulting in rising concentrations, and therefore, upregulation of their respective receptors. When the period of heat passed, this robot would immediately seek to correct these deficits regardless of how small they were. The second, alternative behavior that the robot developed (in 3 runs) was to simply stay near the energy source at all times, except when the occasional need to repair arose; this behavior permitted the robot to continue consuming energy during the hot temperature climate period, with only very limited movement needed. This behavior has a similar basis to the hibernation behavior observed, but was linked to upregulation of T1 and E1 receptors, i.e., there was no upregulation of the H1 receptor. The behavior adopted (hibernation or staying near) depended on the robot's early life experiences, and specifically on the amount of health loss that occurred during the first few cycles. If a robot completed these cycles with minimal health loss, then the second behavioral phenotype was adopted. If on the other hand a robot had health deficits (from early collisions), it adopted the hibernation behavior. Interestingly, it appeared that after the initial cycles and the adoption of one of these two behaviors, future health loss (or lack of health loss) did not result in the robot changing phenotype.

As shown in Fig. 8, the epigenetic robot explored a large portion of the environment while maintaining a low temperature. In contrast, the two other robots moved primarily between the two resources located in the top left and bottom right corners but had high internal temperatures. For the hormone-modulated robot, these high temperature periods were normally a result of running low on energy and needing to move quickly to replenish. For the basic architecture robot, however, these periods simply represented a lack of adaptation to the increased ambient temperature, as it continued its normal behavior even in the hot conditions.

E. Results of Experiment 5: Uneven Resources

Results are shown in Fig. 9. The basic architecture model was in all cases able to operate at a low but passable level.

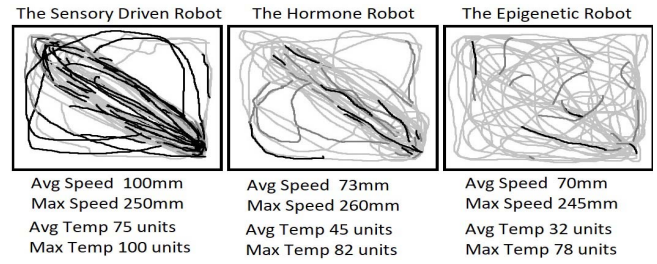


Fig. 8. Movement maps of the three different robots. The color of the lines represents their internal temperature: light gray a temperature between 0–50 units, dark gray 51–75 units, and black 76–100 units.

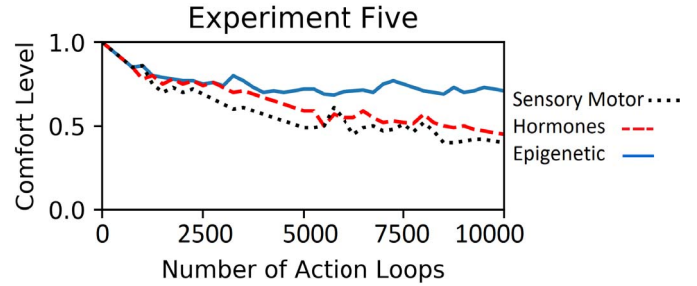


Fig. 9. Results of the fifth experiment (uneven resources).

While there were four deaths spread across the experiments and cases of low comfort levels, this was the only time, excluding the first experiment (Section III-A), when a majority of the basic architecture robots survived. Behaviorally, there is little to report, as it behaved almost exactly the same as in experiment one.

The hormone-modulated model performed better, but still had a hard time dealing with uneven resources. The increased sensory cue from the abundant resources meant that the homeostatic deficit correlating to the rare resources was on average 32% greater. On occasions when the rare resource was hidden behind another resource, the robot would stop searching for it, as it became distracted by other needs.

In contrast, the epigenetic model was able to successfully adapt. Early homeostatic deficits led to the rapid secretion of hormones triggering early epigenetic changes and increasing sensitivity of the associated hormone receptors. This in turn made the robot more sensitive to these deficits, filtering out the distraction and noise of the other resources. When both stimuli were present, the deficit of the overabundant resource would need to be around 42% greater in order to attract the robot's attention. Further, in the four cases when the rare resource was hidden, the deficit related to the common resource would need to drop to between 60% and 80% before the search for the rare resource was interrupted.

Finally, the set of experiments with the high level of temperature (set 3) did not yield considerable differences in terms of homeostatic comfort levels between the three models. While the epigenetic model performed better in the experiments, the biggest difference was qualitative in terms of the behaviors that emerged from the models. The epigenetic model developed a "stalking-like" approach due to upregulation of the T1 and E1 receptors. This behavior emerged in a similar manner to the pounce behavior seen in the third experiment

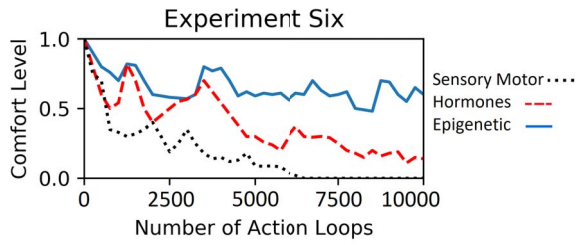


Fig. 10. Results of the sixth experiment (temporal dynamics).

(Section III-C) except that the degree of upregulation was smaller (see Table IV). This meant that movement was not completely suppressed and the robot displayed a very slow exploration behavior until a resource was detected, at which point it moved quickly. In contrast, the hormone-modulated model would maintain a slow constant speed due to equal levels of the E1 and T1 and move toward the desired resource. While both behaviors worked, the approach of the epigenetic robot generated less overall heat, and was therefore slightly preferable for this environment.

F. Results of Experiment 6: Temporal Dynamics

As can be seen in Fig. 10, the basic architecture model was unable to deal with the temporal dynamics, never managing to survive a full run. In the first five sets of runs, when the resource was available for 30 s, the robot was able to survive around eight minutes on average using the same behavior as in experiment one (Section III-A). However, as the period for which the temporal resource was available decreased, the robot increasingly struggled and by the final five sets of runs all agents had died before the 2-min mark. The main problems for this robot were its inability to adapt to “rare” conditions, as in experiment five (Section III-E), and the inability to adapt its speed in order to move promptly toward the temporal resource. Overall, when a temporal resource was available, the basic architecture model managed to reach it only 17% of the time on average. In contrast, the average success rate was 84% for the epigenetic and 62% for the hormone-modulated models.

Comparing the epigenetic and hormone-modulated models proved more interesting. Both robot types performed at a similar level during the first five runs with the 30-s window of opportunity. While the epigenetic robot moved more promptly to resources when they appeared due some upregulation of the E1 receptor, neither robot was ever in any real danger of missing an opportunity. For the epigenetic robot, upregulation of the E1 receptor occurred as a result of increased energy deficits and the resultant high E1 hormone concentrations, due to the fact that the resource was not always available when needed. However, as the window of opportunity shrank, the differences between the two models became apparent. Since the point where the resource would appear next was unknown to the robots, it was inevitable that both architectures would now miss some opportunities to replenish, although the epigenetic robot was generally quicker to move to the energy resource due to heightened sensitivity as a result of the upregulation of the E1 receptor. Finally, due to missed opportunities to fully recover from deficits, both robots often had high levels of the

D1 hormone. This in turn resulted in increased aggressive-like behavior and the occurrences of collisions in later runs, increasing the need for the repair resources. In multiple cases this lead to similar or greater deficits in health in comparison to energy, and therefore, to similar or greater hormone concentrations of H1 in comparison to E1. As health deficits could be, and were readily recovered in the epigenetic robot, they did not lead to any significant upregulation of the H1 receptor. However, the health deficits did cause the hormone-modulated robot to sometimes go to the readily available repair resource during the limited periods when the energy source was present and seen. This was due to the similar effects that both hormone concentrations had on this robot. In contrast, the epigenetic model, due to the upregulation and heightened sensitivity of the E1 receptor had effectively adapted to the rarity of the resource. Even with health deficits present, it only missed the opportunity to replenish energy once, when its health levels were critical (both under 10% and at least 63% less than energy). In total, seven of the hormone-modulated robot runs ended prematurely compared to a single death in the epigenetic model.

IV. CONCLUSION

Drawing inspiration from development in biological organisms, an epigenetic process in which environmental stimuli are able to affect hormone receptors could potentially provide a very useful general mechanism for adaptation in autonomous robots. We have implemented a simple model of such an epigenetic mechanism, in which the sensitivity of hormone receptors is directly regulated by the hormone level, and our results show that it successfully leads to rapid adaptation to different environmental challenges. Our experiments show that this epigenetic mechanism significantly and consistently improves the robot’s adaptability in comparison to a basic homeostatically controlled motivated architecture, and an improved version of this basic architecture that uses (simulated) hormonal modulation. In all our experiments, the epigenetic robot proved better at maintaining homeostasis than the other models. In addition, the variance between the runs in experimental sets of the epigenetic robot was considerably lower, indicating more consistent performance. This improved adaptability resulted from changes in the regulation of hormone receptors that in turn changed the robot’s sensitivity to the variations of the simulated hormones, which signaled to the robot both its internal state and its surrounding environment. These changes to receptors led to the development of different tolerances and sensitivities to the internal and external conditions of the robot (for example, in an environment with plentiful energy supply the robot would tolerate larger energy deficits) that resulted in suitable adaptation of the robot to different environments, shown in both appropriate (and measurable) regulation of its internal survival-related variables and the emergence of successful functional behaviors and tactics such as pushing or hibernation. In each case, at the start of the experiments identical robots (i.e., robots with similar internal parameter values) were used, and the described adaptive qualities and behaviors

were achieved “in real time” and without the need to fine-tune the architecture for the experiments.

The versatile and robust adaptation capabilities shown by our epigenetic architecture in the six environments where it was tested, which posed very different challenges for the robot, supports our view that an epigenetic mechanism can provide a very useful general mechanism for adaptation in autonomous robots. This is corroborated by the application of our epigenetic model to other contexts, such as human-robot interaction [16], [24], formation of group structures [25], and coupled to a neural network to modulate both the learning and behavior of a robot [16].

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